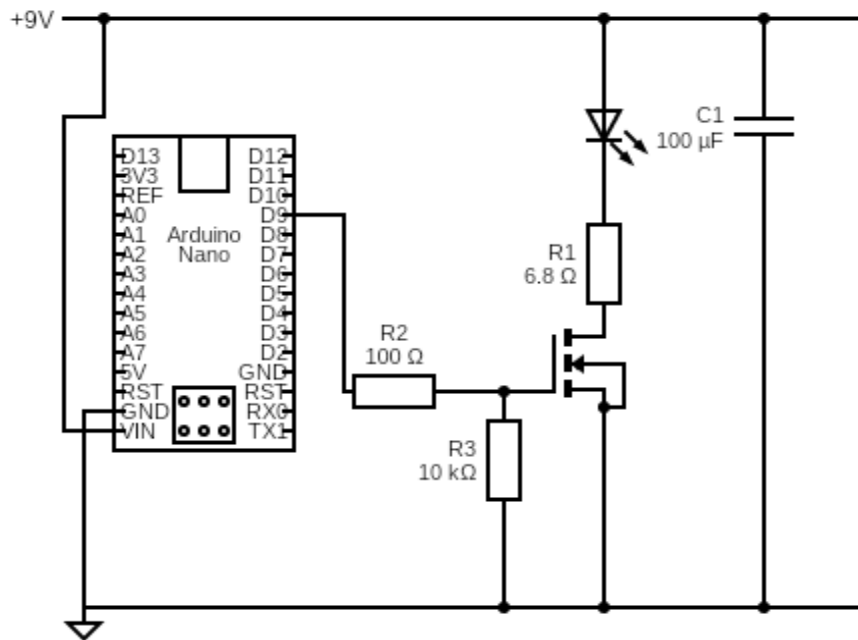


Speed update on UVC LED communication



Electrical diagram of LED drive setup

Arduino: PWM on D9 for LED, powered via vin, +9VDC supply

R2(100Ω): protects Arduino from current spikes when the MOSFET is charging/discharging its gate capacitance

R3(10kΩ): pull-down resistor from MOSFET gate to GND

MOSFET: fast switch for the LED

R1(68Ω, 2W): sets the LED's current to 350mA via Ohm's law

C1(47μF): local energy reservoir near the LED

Math:

1. Needed gate current for 1MHz speed (50ns) -

$$I_{gate} = \frac{Q_{gate}}{t_{rise}} = \frac{15nC}{50ns} = 300mA$$

- For slower current setup goal of at least 100kHz:

$$I_{gate} = \frac{Q_{gate}}{t_{rise}} = \frac{15nC}{1000ns} = 15mA$$

- For higher speeds a gate driver might be needed

2. R1 -

$$R_1 = \frac{V_{supply} - V_f - V_{ds on}}{I_f} = \frac{9V - 6.5V - 0.1V}{0.350A} = 6.86\Omega$$

- Standard E12 value: 6.8Ω, which gives 0.353A (within ±1% of target - acceptable)

$$P_{R1 peak} = I_f^2 R_1 = 0.353^2 \times 6.8 = 0.847W$$

- At 50% duty is 0.42W
- Component rating chosen: 2W

3. Supply voltage -

If V_f drifts by ΔV_f , I changes by:

$$\Delta I = \frac{\Delta V_f}{R_1}$$

Current error:

$$\frac{\Delta I}{I_f} = \frac{\Delta V_f}{V_{R1}}$$

Supply	V_{R1}	ΔI for 100mV shift	Power wasted in R1
7.5V	0.9V	11%	0.16W
9V	2.4V	4.2%	0.42W
12V	5.4V	1.9%	1.89W

- Choice: 9V

4. R2 -

$$I_{peak GPIO} = \frac{V_{GPIO}}{R_{GPIO} + R_2}, R_{GPIO} \approx 25\Omega$$

- For $I = 40mA$ we need $R_2 = 100\Omega$

5. R3 -

Pull-down, so 10kΩ is standard

6. C1 -

$$C \geq \frac{I_f t_{on}}{\Delta V_{allowed}} = \frac{0.353 \times 5 \times 10^{-6}}{0.05} = 35 \mu F$$

- Choice: 100 μF aluminum polymer

7. Thermal management -

$$P_{thermal} = V_f I_{avg} (1 - \eta) = 6.5 \times 0.175 \times (1 - 0.03) = 1.10 W$$

η - efficiency (3%)

- Heat flow equation:

$$T_j = T_a + P_{thermal} (R_{\theta JC} + R_{\theta CS} + R_{\theta SA}),$$

- $R_{\theta JC} \approx 10^\circ C/W$ (LED datasheet)
- $R_{\theta CS} \approx 1^\circ C/W$ (thermal interface material)
- $R_{\theta SA}$ - depends on heatsink

- Max $R_{\theta SA}$ for $T_j \leq 70^\circ C$, $T_a = 25^\circ C$:

$$R = \frac{T_j - T_a}{P_{thermal}} - (R_{\theta JC} + R_{\theta CS}) = \frac{70 - 25}{1.10} - 10 - 1 = 30^\circ C/W$$

A current solution is a heatsink with these dimensions:

- Overall: 50 × 25 × 10 mm
- Base thickness: 1.5 mm
- Pin array: 8 × 9 = 72 pins
- Pin dimensions: 4 × 1.15 mm cross-section, height = 10 - 1.5 = **8.5 mm**
- Anodized black

The surface area of this heatsink is roughly 96.8cm², when we multiply by 0.7 efficiency factor we get 67.8cm². The estimated $R_{\theta SA}$ then is:

$$R_{\theta SA} = \frac{650}{67.8} = 9.6^\circ C/W$$

Round up for conservatism (real-world performance is usually slightly worse than the formula):

$R_{\theta SA} \approx 10\text{-}12^\circ C/W$ in still air.

BOM:

R2(100Ω)	x1
R3(10kΩ)	x4
MOSFET	at least 2
R1(68Ω, 2W)	x1
C1(100μF)	x1
100 nF X7R ceramic	x2
Cf(16 pF C0G/NP0 ceramic)	x1
Rf (1 MΩ, 1/4 W)	x1
OPA380AID - Photocurrent amplifier	x1
SOIC-to-DIP adapter	x1
LM393N	x1
SG01D-5Lens SiC UV Photodiode	x1 (не го намирам в mouser + всички фотодиоди са мега скъпи)

Brief explanation of sensor parts:

Rf (1 MΩ) — sits between the OPA380's output and its inverting input. It's how the TIA converts current to voltage. The bigger Rf is, the more voltage you get per nanoamp of photocurrent. 1 MΩ gives you 1 V per microamp.

Cf (16 pF) — sits in parallel with Rf. Without it, the TIA oscillates because the photodiode's own capacitance creates a phase shift that turns negative feedback into positive feedback. Cf damps this and also sets the upper bandwidth (1 MHz).

Two 10 kΩ resistors as voltage divider — split your 5 V supply into a 2.5 V reference. This 2.5 V goes to the LM393's inverting input as the threshold. When the TIA output is above 2.5 V, the comparator outputs HIGH; below, LOW.

Third 10 kΩ resistor as pullup — the LM393 has an "open-collector" output, meaning when it wants to output HIGH it doesn't actually drive 5 V — it just disconnects, leaving the output floating. The pullup resistor pulls the output up to 5 V whenever the comparator releases it. Without this resistor, the Arduino sees a floating pin instead of a clean HIGH.

Two 100 nF capacitors — one connected from the OPA380's power pin to ground, one from the LM393's power pin to ground. They keep the supply voltage stable when the chips draw sudden current spikes (which they do constantly during fast switching). Without bypass caps, chips behave unreliably and pick up noise.

OPA380AID - TIA converts current (which the Arduino can't read at all) into voltage (which it can read, but slowly)

LM393N - Comparator converts slow-to-read voltage into fast-to-read digital pulses (which the Arduino can read in microseconds)

See diagram below:

